THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 3

Chapters 9–12: **Light - Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Resistors of 2 ohm, 3 ohm and 6 ohm are connected in parallel. The equivalent resistance is 1

(a) 11 ohm **(b)** 2 ohm **(c)** 1 ohm **(d)** 3 ohm

2. A conductor of length L carrying current I is placed parallel to the field. The force is 1

(a) zero **(b)** half of maximum

(c) maximum **(d)** equal to the weight

3. A convex lens of f = 15 cm forms a real image at 60 cm. The object distance is 1

(a) 15 cm **(b)** 45 cm **(c)** 30 cm **(d)** 20 cm

4. Fine particles in the atmosphere scatter mainly 1

(a) blue light **(b)** no light **(c)** yellow light **(d)** red light

5. A 2 kW heater runs for 30 minutes. Energy consumed is 1

(a) 1 kWh **(b)** 60 kWh **(c)** 2 kWh **(d)** 4 kWh

6. A coil with 100 turns instead of 50 turns, other things same, gives 1

(a) the same induced emf **(b)** a larger induced emf

(c) no emf **(d)** a smaller induced emf

7. A convex mirror has f = 20 cm. An object is at 30 cm. The image distance is 1

(a) 12 cm in front **(b)** 12 cm behind the mirror

(c) 20 cm behind **(d)** 60 cm behind the mirror

8. A myopic person's eyeball is too long. His eye forms the image of a distant object 1

(a) in front of the retina **(b)** on the retina

(c) at infinity **(d)** behind the retina

9. The V-I graph of a conductor has a slope of 5 ohm. If the current is 2 A, the voltage is 1

(a) 7 V **(b)** 2.5 V **(c)** 0.4 V **(d)** 10 V

10. When current in a solenoid is doubled (other factors same), the field inside 1

(a) remains the same **(b)** increases **(c)** decreases **(d)** becomes zero

11. The refractive index of glass is 1.5 and of water is 1.33. Light travels fastest in 1

(a) water **(b)** glass **(c)** both equally **(d)** neither

12. When a second inverted prism is placed in the path of dispersed light, we get 1

(a) only red **(b)** more colours **(c)** no light **(d)** white light

13. Two resistors of 4 ohm and 12 ohm in series are connected to a 16 V battery. The current is 1

(a) 1 A **(b)** 16 A **(c)** 0.25 A **(d)** 4 A

14. Which of the following reverses the force on a current-carrying conductor in a field? 1

(a) Heating it **(b)** Reversing the current

(c) Increasing the current **(d)** Reducing the length

15. The magnification of a lens is +3. The image is 1

(a) virtual, erect and three times larger **(b)** real and magnified

(c) virtual and diminished **(d)** real and inverted

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why is the series arrangement not used for home appliances? 2

17. How many times does the current of frequency 50 Hz change direction in 1 minute? 2

18. Find the magnification when a convex mirror of f = 15 cm forms an image at 7.5 cm behind it of an object at 15 cm. 2

19. A person can see clearly only up to 2 m. Find the power of the lens to correct this. 2

20. A 10 ohm resistor and a 15 ohm resistor are connected in series to a 5 V cell. Find the current. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Find the equivalent resistance of 3 resistors of 4 ohm, 6 ohm and 12 ohm connected in parallel and the current drawn from 6 V. 3

22. How does the magnetic field due to a solenoid vary 3

(a) with the current

(b) with the number of turns per unit length

(c) with a soft iron core?

23. A 4 cm high object is placed 12 cm from a convex lens of f = 8 cm. Find the image height and nature. 3

24. Explain how a prism in the inverted position recombines the spectrum. 3

OR

Why do two bulbs of 60 W and 100 W, both rated 220 V, have different resistances? Which has more? 3

25. Why does the induced current in a generator last only during rotation? 3

OR

State what will happen to the focal length of a lens if it is dipped in water. Explain. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. A convex mirror has a radius of curvature of 40 cm. An object 5 cm tall is 20 cm from it. Find the image distance, size and nature. 5

OR

A short-sighted person has a far point at 40 cm. 5

(a) Name the defect

(b) find the power of the lens to see far objects

(c) explain why the lens is concave.

27. Three resistors of 10 ohm, 20 ohm and 30 ohm are in series. 5

(a) Find the equivalent resistance.

(b) What is the current if a 12 V battery is used?

(c) Find the power dissipated in each.

OR

A student wants to make a strong electromagnet. Describe how it is made, state the factors on which its strength depends and state why it is called a temporary magnet. 5

END OF QUESTION PAPER **Total: 50 Marks**

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